Chapter 57. Sleep in the Ventilator-Supported Patient

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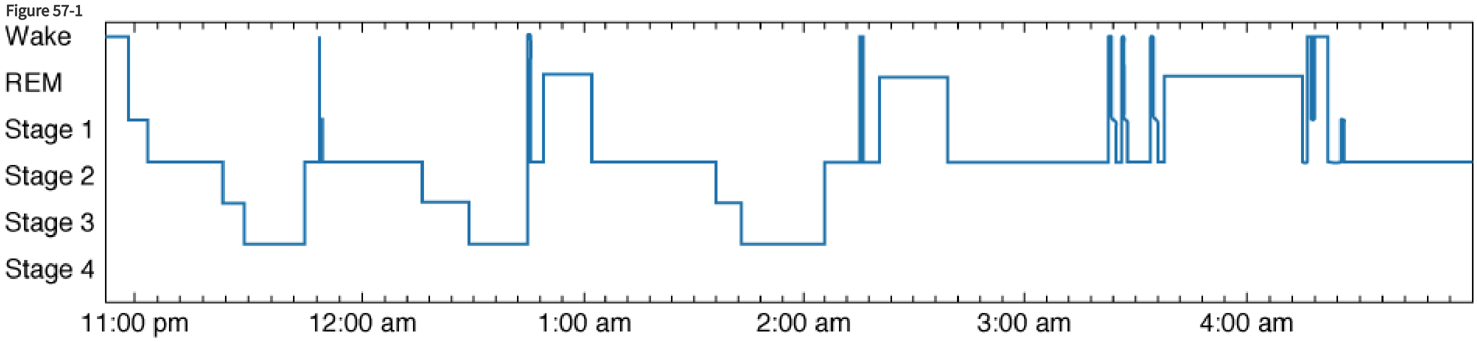
Sleep in the Ventilator-Supported Patient: Introduction

It is well recognized that sleep is abnormal in mechanically ventilated patients in the intensive care unit (ICU). Although this has been described for decades, there is still no consensus on the underlying pathogenesis and the best way to manage it. Moreover, the assumption that abnormal sleep is not good for patients who are critically ill is based primarily on extrapolation from models of sleep loss and sleep disruption in other patient populations and not on evidence that abnormal sleep affects the clinical outcomes of patients in the ICU. Nevertheless, there is growing interest in this topic as the technology to measure sleep evolves and new ways are sought to improve patients' ability to recover from their critical illness. This chapter outlines the current understanding of the causes and potential consequences of sleep disruption in ventilator-supported patients and how this may be further researched and treated.

Normal Sleep

Sleep is objectively assessed by means of polysomnography, the simultaneous recording of several electroencephalographic and physiologic parameters.[1] Sleep periods are classified as non–rapid eye movement (NREM) and rapid eye movement (REM) sleep. NREM sleep is further subdivided into four stages, with stages 3 and 4 also referred to as slow-wave sleep (SWS). Each sleep stage is recognized by characteristic changes on the electroencephalogram and, in addition, REM sleep has distinctive, intermittent rapid eye movements. During normal sleep, periods of NREM and REM alternate throughout the night in a recognizable pattern, so that most SWS occurs during the first half of the night and most REM sleep occurs during the second half (Fig. 57-1). The "normal" duration of sleep required and the proportion of time spent in each stage of sleep depends on many factors including age and genotype.[2] In healthy, middle-aged individuals, however, nocturnal sleep lasts 7 to 8 hours, and 5% to 10% of that time is spent in stage 1 NREM sleep, 50% in stage 2 NREM sleep, 15% to 20% in SWS, and 25% in REM sleep.[1] There are also standardized electroencephalographic criteria for identifying arousals and awakenings on the polysomnograph;[3,4] up to 10 arousals per hour of sleep is considered to be within normal limits.[5] The term *sleep architecture* refers to the amount of time spent in each sleep stage and *sleep disruption* is reflected by an increased frequency of arousals and awakenings.

Figure 57-1



Normal sleep (nocturnal hypnogram). *Vertical axis: REM,* rapid eye movement sleep; *NREM,* non–rapid eye movement sleep stages 1, 2, 3, and 4. *Horizontal axis:* Time in hours.

Sleep in the Intensive Care Unit

Patient Perception

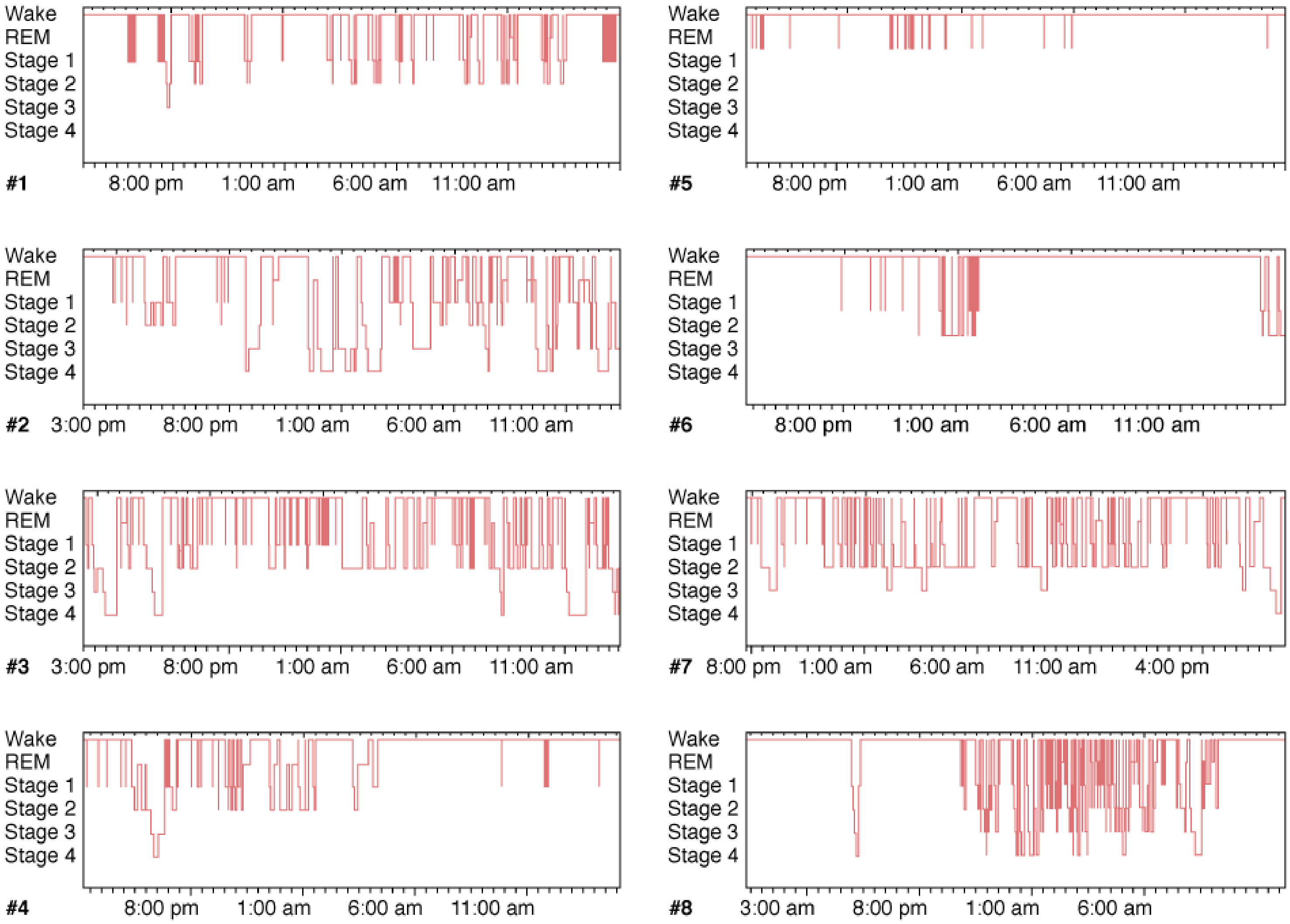
When questioned after discharge from the ICU, patients consistently report sleep disruption during their stay.[6–11] In one study,[10] 200 patients from four different ICUs received questionnaires that evaluated the quality of their sleep at home and in the ICU and the effect of noise and a variety of activities (such as nursing interventions, phlebotomy, and diagnostic procedures) on their sleep quality. Sleep quality in the ICU was perceived as significantly poorer than sleep quality at home. In addition, sleep quality did not change significantly over the course of the ICU admission, and no differences were reported between ICUs or between ventilated and nonventilated patients. Ventilated patients, however, reported greater daytime sleepiness, perhaps because of the greater administration of sedatives or more severe illness. Patients selected the recording of vital signs and phlebotomy as the most sleep-disrupting environmental factors. Noise was not rated as the primary cause of sleep disruption; of the many forms of environmental noise stimuli, however, communication between staff members and telemetry alarms were perceived as the most disruptive. Although very thorough in design and statistical analysis, the study was limited by potential recall bias, lack of objective sleep assessment by polysomnography, and the absence of a control group.

Polysomnography

Since the mid-1970s, numerous polysomnographic studies in ICU patients have consistently revealed both sleep fragmentation and sleep loss.[6,12–19] Hilton[6] studied nonventilated patients by 24-hour polysomnography and found a decreased total sleep time, an increase in stage 1 NREM sleep, and a concomitant decrease in SWS and REM sleep. Hilton also observed an apparent uncoupling of sleep from the day–night circadian pattern: Only 50% to 60% of sleep occurred at night.[6] These characteristic sleep patterns have also been observed in subsequent 24-hour polysomnography studies.[12,16,20]

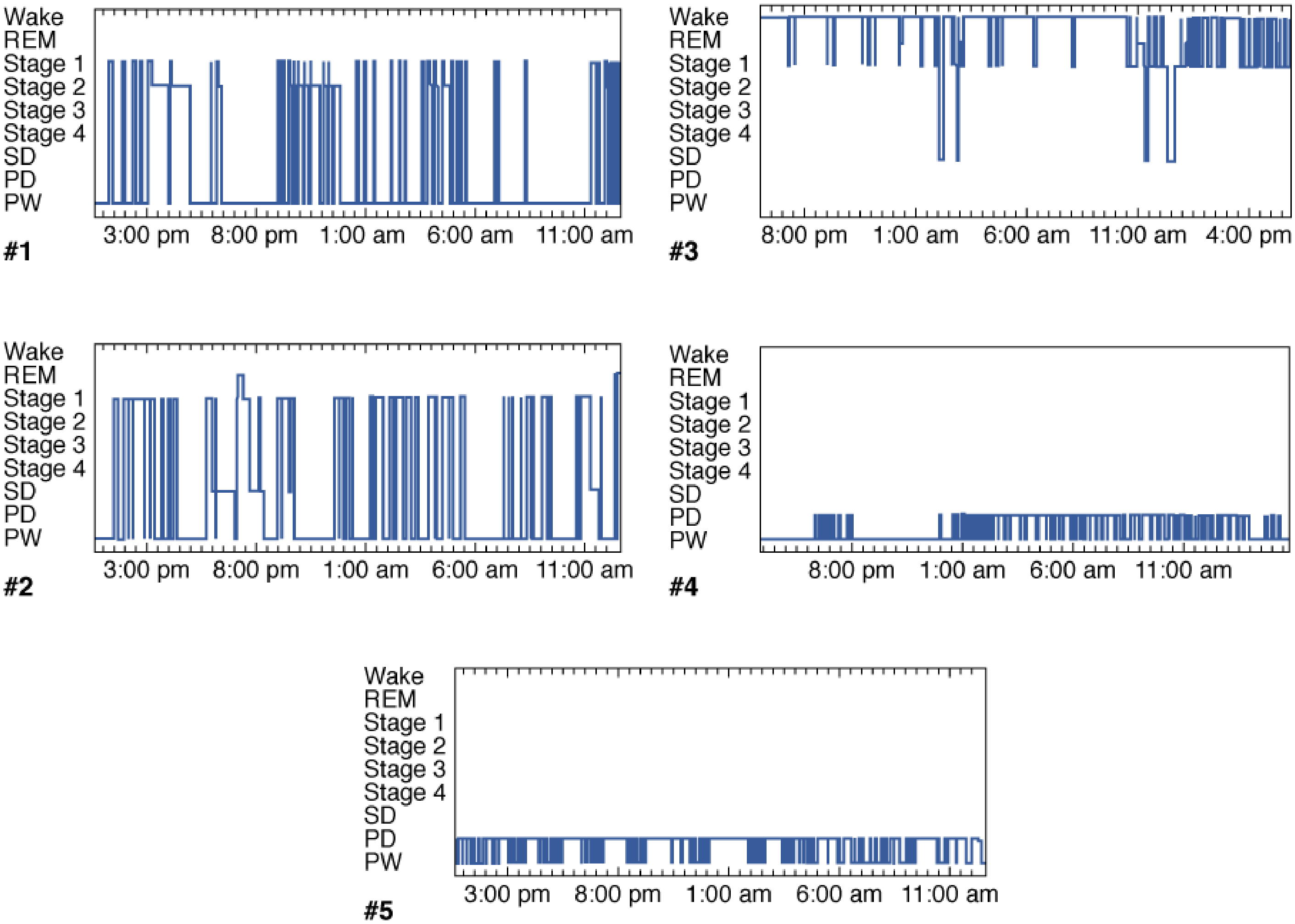
Cooper et al investigated sleep in ventilated patients in the ICU and categorized patients into three groups based on polysomnographic findings: disrupted sleep, atypical sleep, and coma.[12] As seen in other ICU cohorts, patients in the disrupted sleep group (Fig. 57-2) showed the abnormal temporal distribution of sleep described earlier as well as reduced amounts of SWS and REM sleep and an increased frequency of arousals and awakenings. Patients in the atypical sleep group had electroencephalogram (EEG) features intermediate between sleep and coma, characterized by a virtual absence of stage 2 NREM sleep and REM sleep (Fig. 57-3). In addition, patients displayed “pathologic wakefulness,” where behavioral correlates of wakefulness (such as saccadic eye movements and sustained chin muscle activity) coincided with EEG features of SWS (Fig. 57-4). The coma group was characterized by EEG features of coma according to the classification of Young et al.[21] The authors concluded that sleep could not be identified by polysomnography in all critically ill patients. They proposed the following criteria to select ICU patients in whom sleep can be reliably measured by polysomnography: acute physiology score less than 13, Glasgow Coma Scale score greater than 10, and sedative doses of lorazepam equivalents and morphine equivalents less than 10 mcg/kg per hour. These cutoffs approximated the point estimate of the atypical sleep group. Subsequent data suggest that sleep disruption can be measured by polysomnography in approximately 50% of patients in a general ICU.[22]

Figure 57-2



Disrupted sleep in the ICU (24-hour hypnograms in eight patients). *Vertical axis: REM,* rapid eye movement sleep; *NREM,* non–rapid eye movement sleep stages 1, 2, 3, and 4. *Horizontal axis:* Time in hours. Note (1) sleep was distributed throughout the 24-hour period in all patients except patient 8, in whom sleep was predominantly nocturnal; (2) frequent awakenings; and (3) prolonged wakefulness, especially patients 4, 5, and 6. (Reproduced, with permission, from the American College of Chest Physicians, from Cooper AB, Thornley KS, Young GB, et al. Sleep in critically ill patients requiring mechanical ventilation. *Chest.* 2000;117:809–818.)

Figure 57-3



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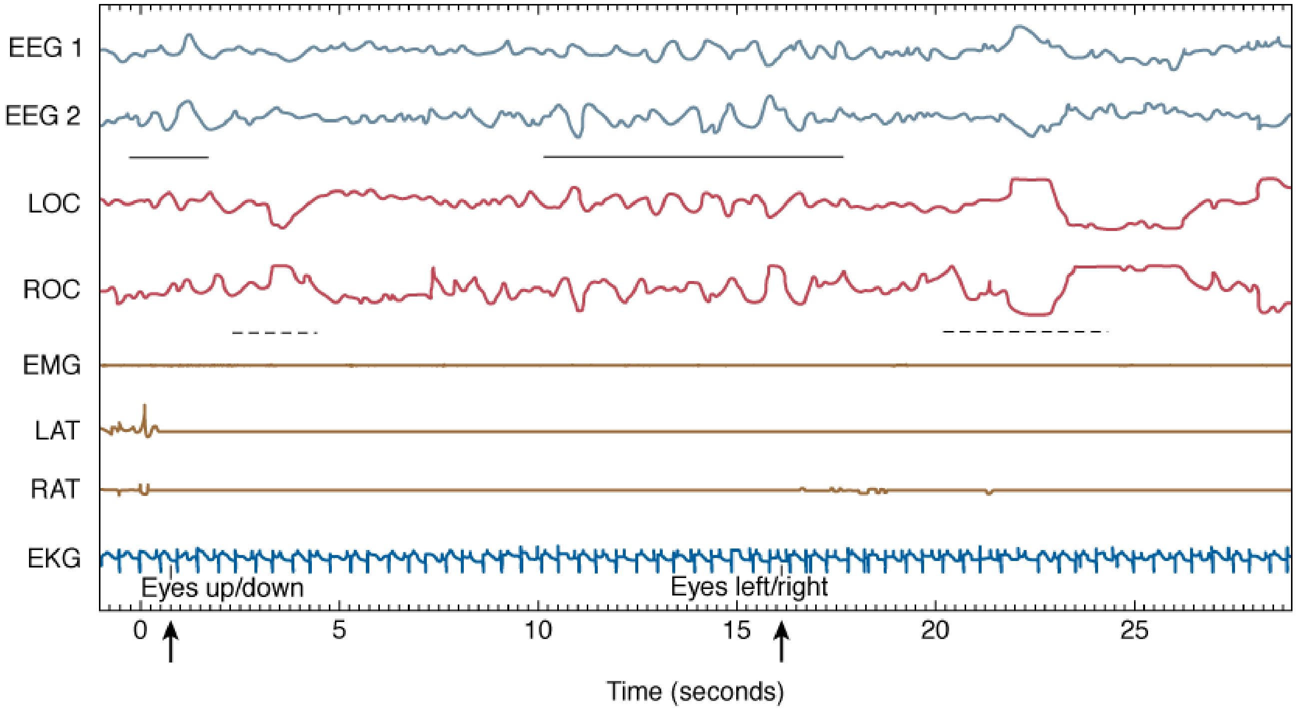
Atypical sleep in the ICU (24-hour hypnograms in five patients). *Vertical axis: REM,* rapid eye movement sleep; *NREM,* non–rapid eye movement sleep stages 1, 2, 3, and 4; sleep delta (*SD*); pathologic delta (*PD*); and pathologic wakefulness (*PW*). *Horizontal axis:* Time in hours. (Reproduced, with permission, from the American College of Chest Physicians, from Cooper AB, Thornley KS, Young GB, et al. Sleep in critically ill patients requiring mechanical ventilation. *Chest.* 2000;117:809–818.)

Figure 57-4



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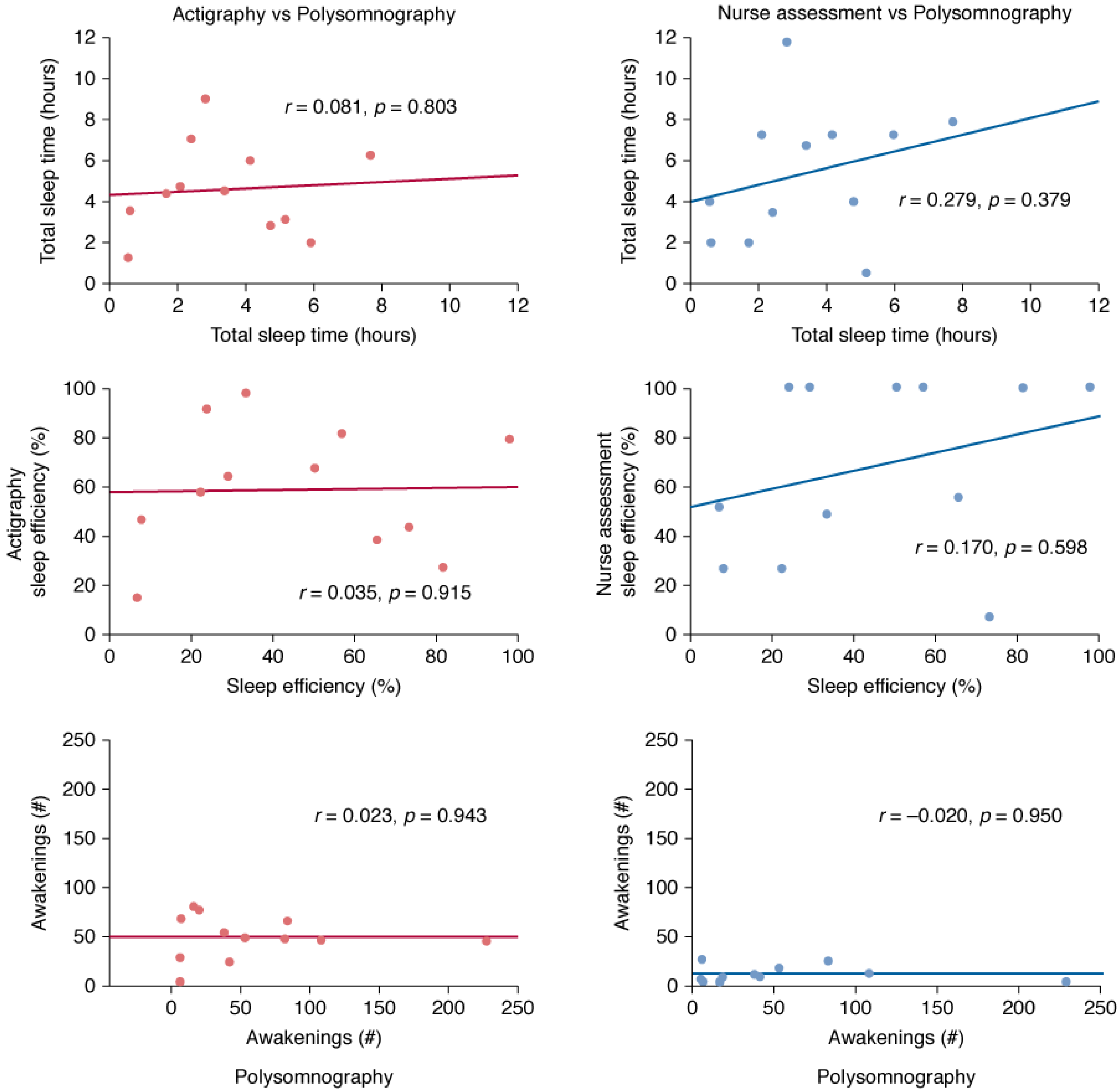
Pathologic wakefulness during atypical sleep (30-second epoch). *Vertical axis: EEG,* electroencephalogram; *EKG,* electrocardiogram; *EMG,* submental electromyogram; *LAT,* left anterior tibialis EMG; *LOC,* left oculogram; *RAT,* right anterior tibialis EMG; *ROC,* right oculogram. *Horizontal axis:* Time in seconds. Note the slow-wave EEG activity (indicated by the *solid horizontal bar*) during the patient's responses (eye movements, indicated by *broken horizontal lines*) to biocalibration (indicated by a *bold arrow*). (Reproduced, with permission, from the American College of Chest Physicians. Cooper AB, Thornley KS, Young GB, et al. Sleep in critically ill patients requiring mechanical ventilation. *Chest.* 2000;117:809–818.)

Once polysomnography has been performed, what is the most reliable way to score sleep in the ventilator-supported patient? Sleep is conventionally assessed by manual scoring of the EEG according to the criteria of Rechtschaffen and Kales.[4] The reliability, however, of this methodology to score sleep in the critically ill patient has been questioned.[23] Ambrogio et al compared manual scoring of sleep in fourteen ventilator-supported patients in the ICU and seventeen age-matched ambulatory patients in the sleep laboratory and reported that interobserver and intraobserver reliability was weaker for critically ill patients (k = 0.52 ± 0.23) than for those in the sleep laboratory (k = 0.89 ± 0.13; P = 0.03). Furthermore, they suggested that computer-based spectral analysis of the EEG with fast Fourier transformation is a better way to monitor sleep in this patient population. It appears that improved and innovative analysis of the EEG holds the best promise for further progress, particularly because less-invasive methodologies to monitor sleep such as assessment by the bedside nurse and actigraphy, are unreliable (Fig. 57-5).[24–26]

Figure 57-5



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Total sleep time (total duration of sleep during the recording period), sleep efficiency (total sleep time expressed as a percentage of total recording time), and the number of awakenings during the recording period estimated by actigraphy (customized setting shown) and nurse assessment compared with polysomnographic findings. (Used, with permission, from Beecroft et al.[26])

Causes of Sleep Disruption in the Intensive Care Unit

Medical Disorders

Patients may enter the ICU with preexisting medical or sleep disorders, such as asthma or sleep apnea, which cause disruption of sleep if inadequately controlled. More importantly, the acute illness that precipitated the ICU admission, such as major surgery, can disrupt sleep. A very consistent finding in surgical patients is the reduction or absence of both SWS and REM sleep in the immediate postoperative period. This is characteristically followed by “REM rebound,” an increase in both the number of phasic eye movements and the overall amount of

REM sleep.[13,15,18,19] REM rebound may result from the withdrawal of REM-suppressing medications, such as narcotics,[18] analgesics, and benzodiazepines,[27] and/or a decrease in illness-related sleep disruptors such as pain.

Medications

Several medications can alter sleep. A comprehensive review of this topic has been published.[27–29] Table 57-1 summarizes the effects of some medications that are commonly used in the ICU. Although benzodiazepines can increase total sleep time, they are known to reduce both SWS and REM sleep and are associated with rebound insomnia once they are stopped.[30] Opiates also decrease SWS and REM sleep, increase the amount of wakefulness after sleep onset, and are associated with withdrawal hypersomnolence.[27,30,31] Glucocorticoids are often associated with insomnia[29] and many selective serotonin reuptake inhibitors increase alertness and restlessness, in addition to exacerbating sleep disorders such as restless leg syndrome and periodic limb movement disorder.[32]

Table 57-1: Effects of Medications on Sleep

Medication	Clinical Effects	Changes on Polysomnography
CNS medications		
Narcotics	Varies with agent; withdrawal hypersomnolence	Acute: ↑WASO, ↓SWS, ↓REM
Benzodiazepines	Sedation, withdrawal insomnia	↑TST, ↓SL, ↓W, ↓SWS, ↓REM
Tricyclic antidepressants	Improve sleep; may be sedating	Generally ↑TST, ↓W, ↓REM
Selective serotonin reuptake inhibitors	May worsen sleep; few daytime complaints	Generally ↓TST, ↑W, ↓REM
Barbiturates	Sedation, withdrawal insomnia	↑TST, ↓W, ↑ ↓SWS, ↓REM
Phenytoin	Sedation	↓SL
Carbamazepine	Sedation	↓SL and ↑TST
Cardiac medications		
β-antagonists	Insomnia, nightmares	↑W, ↓REM
α_2-agonists (clonidine, methyldopa)	Insomnia, nightmares, sedation	↑TST, ↑↓REM
αβ-antagonists	Insomnia, fatigue, somnolence	No studies
Diltiazem	Insomnia, abnormal dreams, sleepiness	No studies
Amiodarone	Insomnia, nightmares	No studies
Other medications		
Aspirin	—	Acute: ↓SWS
Glucocorticoids	Insomnia	↑W, ↓REM
Theophylline	Insomnia	↑W, ↓TST

Abbreviations: CNS, central nervous system; *REM,* rapid eye movement sleep; *SL,* sleep latency; *SWS,* slow-wave sleep; *TST,* total sleep time; *W,* wakefulness; *WASO,* wakefulness after sleep onset.

Adapted, with permission, from Wooten[27] and Schweitzer.[29]

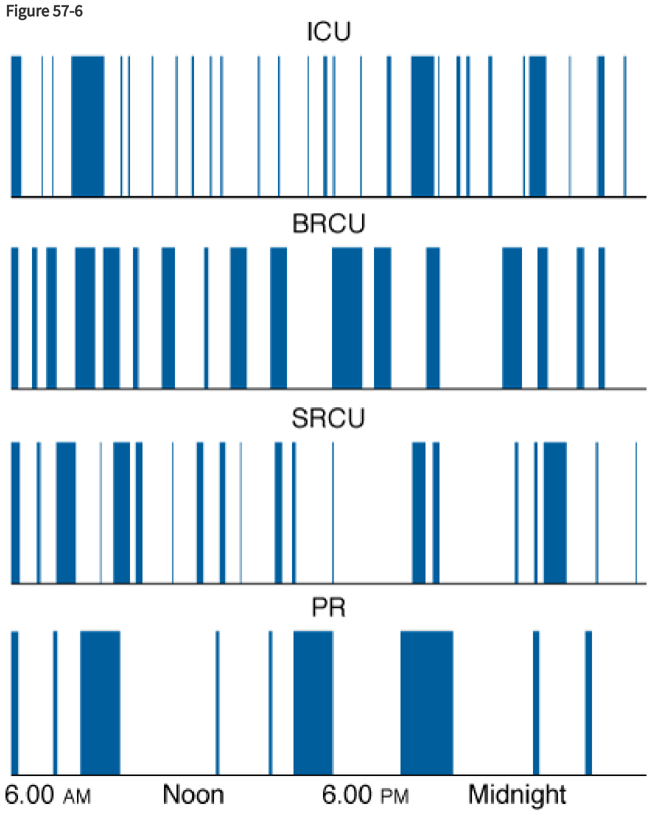
Altered Circadian Rhythm

Almost all biologic functions have a circadian rhythm, which synchronizes interactions both among themselves and with the external environment. Alteration of the circadian rhythm that regulates sleep and wakefulness is a recognized cause of insomnia in patients who are not critically ill[33] and may contribute to sleep disruption in the ICU. The circadian clock has been evaluated in critically ill patients by measuring either core body temperature or melatonin levels in the blood or urine. Two retrospective studies, which measured core body temperature, found that circadian rhythm was absent in 20% to 80% of patients.[34,35] Gazendam et al[36] recorded core body temperature in twenty-one patients and reported that a 24-hour rhythm was detectable in all patients. The rhythm, however, was advanced or delayed by several hours compared with control subjects, and the degree of displacement was correlated with the severity of illness, reflected by Acute Physiology and Chronic Health Evaluation (APACHE) III scores. Several studies have measured melatonin levels in ICU patients.[37–41] Melatonin is secreted by the pineal gland and its release is closely synchronized with sleep in healthy individuals;[42] it starts to rise between 9 and 11 PM, peaks between 1 and 3 AM, and falls to low baseline values between 7 and 9 AM. The characteristic nocturnal rise in melatonin is absent in critically ill patients, and this has been correlated with the use of mechanical ventilation,[40,41] sepsis,[39] and the postoperative period.[38] Although altered circadian rhythm may contribute to sleep disruption in the ICU, this has yet to be proven. Moreover, the extent of its role is likely to vary among individual patients depending on factors such as the ICU environment, illness severity, the length of stay, and the impact of competing sources of sleep disruption.

Intensive Care Unit Environment

Several studies have examined the role of light, patient-care activities, and noise in causing sleep disruption in the ICU.[43–49] Meyer et al[47] found that circadian light levels were maintained in the ICU, and modern ICUs minimize light intensity at nighttime. Consequently, light does not appear to be a significant source of sleep disruption. Nursing interventions occur at least hourly in the ICU[47] (Fig. 57-6), and have been associated with arousals.[13] The presence of excessive noise in the ICU has been thoroughly documented.[43,46–49] Aaron et al observed a strong correlation between the number of sound peaks of greater than 80 A-weighted decibels and arousals from sleep in a group of ICU patients.[48] Balogh et al reported that alarms were the most irritating noise, and observed that even during the night, the longest "quiet" interval was only 22 minutes.[46]

Figure 57-6



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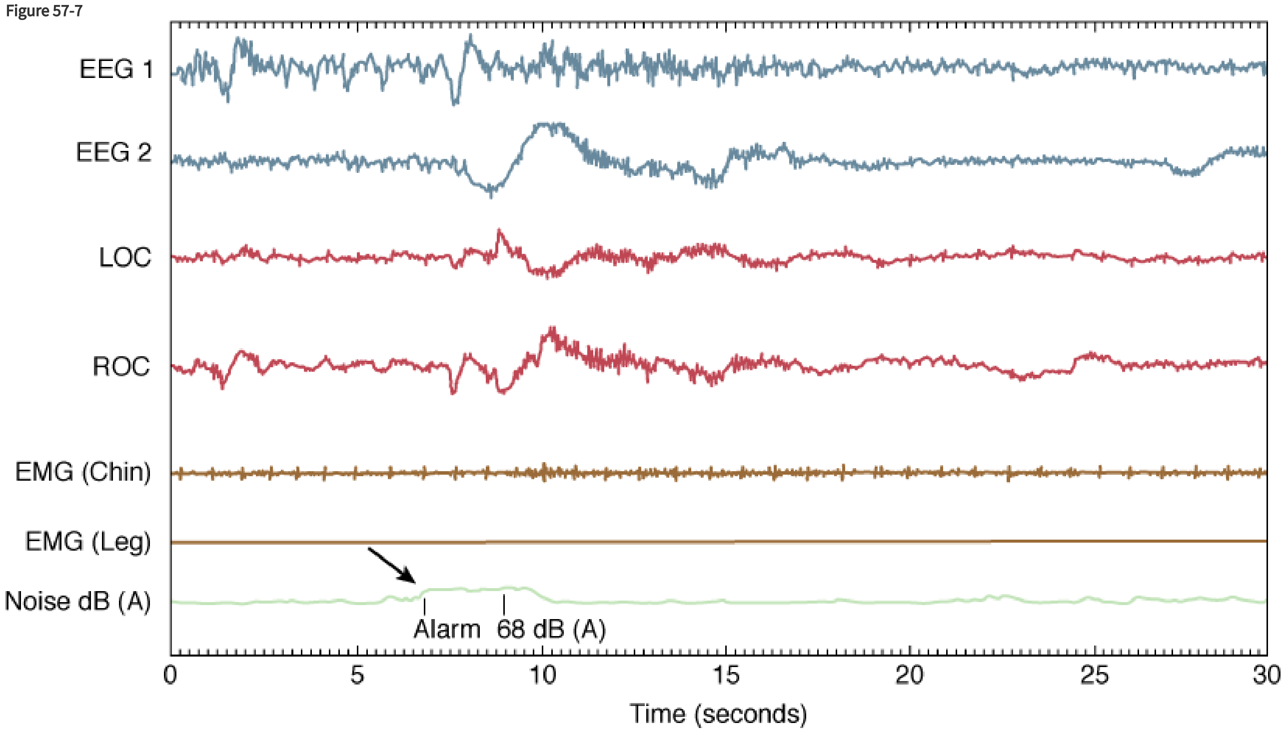
Patient interruptions by staff over 24 hours in four areas: a three-bed medical ICU, a three-bed respiratory care unit (*BRCU*), a single respiratory care unit room (*SRCU*), and a private room (*PR*) on a general medical floor. The *dark areas* represent interruptions and the *clear areas* represent time available for sleep. (Reproduced, with permission, from the American College of Chest Physicians. Meyer TJ, Eveloff SE, Bauer MS, et al. Adverse environmental conditions in the respiratory and medical ICU settings. *Chest.* 1994;105:1211–1216.)

Another investigative approach has been to simulate the noisy ICU environment in a controlled setting such as a sleep laboratory. Exposure of healthy subjects in a sleep laboratory to recorded ICU noise–induced sleep disruption similar to that observed in patients in the

ICU.[44,45,49] These studies in the ICU and sleep research laboratory led to the assumption that sleep disruption in the ICU was predominantly caused by noise and patient-care activities. The ICU studies, however, did not include simultaneous monitoring of noise and arousals. Consequently, the association was, at best, indirect. Furthermore, the simulated ICU environment is limited by the fact that it evaluates the impact of noise in isolation without the interaction of other competing sleep disruptors that are found in the ICU. Consequently, the role of the ICU environment, specifically noise and patient-care activities, were reassessed. Freedman et al,[50] using polysomnography and time-synchronized recording of environmental noise, directly linked noise to arousals. They determined that noise was responsible for only 15% of all arousals and awakenings. Although it was the first study to demonstrate that common noise elevations directly cause arousals in ICU patients, other environmental factors such as patient-care activities were not assessed.

Gabor et al subsequently evaluated the contribution of the ICU environment to sleep disruption in both ventilated patients and healthy subjects, and also evaluated the effectiveness of a noise-reduction strategy (moving the subject from the open ICU to a single room).[51] They performed comprehensive, synchronized monitoring of sleep by polysomnography, noise (calibrated sound meter), and all patient-care activities (audiovisual recording) for 24 hours. Although loud noise and frequent patient-care activities were prevalent in the ICU environment, they were responsible for less than 30% of the observed sleep disruption (Fig. 57-7). Healthy individuals slept relatively well in this potentially disruptive environment. Although noise accounted for a significant proportion of sleep disruption in this group, its extent was not pathologic. A quantitative improvement in sleep quality was observed as a result of noise reduction; however, there was no change in sleep architecture. The cause of 68% of arousals and awakenings in these mechanically ventilated patients could not be attributed to noise or any patient-care activity (Table 57-2).

Figure 57-7



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Polysomnographic example of a noise-induced arousal. *Vertical axis: EEG,* electroencephalogram; *EMG,* electromyogram; *LOC,* left oculogram; *ROC,* right oculogram. *Horizontal axis:* Time in seconds. *Arrow* indicates abrupt increase in noise (68 A-weighted decibels) caused by an alarm, followed by an arousal from stage 2 NREM sleep. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Gabor JY, Cooper AB, Crombach SA, et al. Contribution of the intensive care unit environment to sleep disruption in mechanically ventilated patients and healthy subjects, *Am J Respir Crit Care Med.* 2003;167:708–715. Offical Journal of the American Thoracic Society.)

Table 57-2: Impact of Noise and Patient-Care Activities on Sleep Disruption

Event Type	Number per Hour of Sleep	Percentage Causing Disruption	Percentage of Total Disruption
Sound	36.5 ± 20.1	11.7 ± 8.3	20.9 ± 11.3
Family visits	0.7 ± 0.7	38.6 ± 39.3	1.0 ± 1.3
RT/physio	0.4 ± 0.5	30.7 ± 32.6	0.5 ± 0.7
Suctioning	0.2 ± 0.8	62.5 ± 47.9	0.6 ± 0.8
RN visits	3.5 ± 1.8	21.7 ± 11.6	4.1 ± 3.5
Assess vitals	0.3 ± 0.4	51.4 ± 34.4	0.7 ± 0.9
Med admin	2.7 ± 3.1	49.4 ± 25.6	0.9 ± 1.0
All medical care	7.8 ± 4.2	17.7 ± 5.4	7.1 ± 4.4
Apparatus/tech	1.1 ± 1.0	21.6 ± 26.3	1.4 ± 1.8
Unidentifiable	—	—	68.1 ± 9.7

Abbreviations: Apparatus/tech, noise from the sleep/environmental monitoring equipment or actions by the attending research assistant; *Med admin,* administering medication to a patient; *RT/physio,* any care provided by a respiratory therapist or physiotherapist; *RN visits,* any care provided by a nurse; *Sound,* sound peaks.

Source: Reprinted with permission of the American Thoracic Society. Copyright © 2003 American Thoracic Society. Gabor JY, Cooper AB, Crombach SA, et al. Contribution of the intensive care unit environment to sleep disruption in mechanically ventilated patients and healthy subjects. *Am J Respir Crit Care Med.* 2003;167:708–715. Official Journal of the American Thoracic Society.

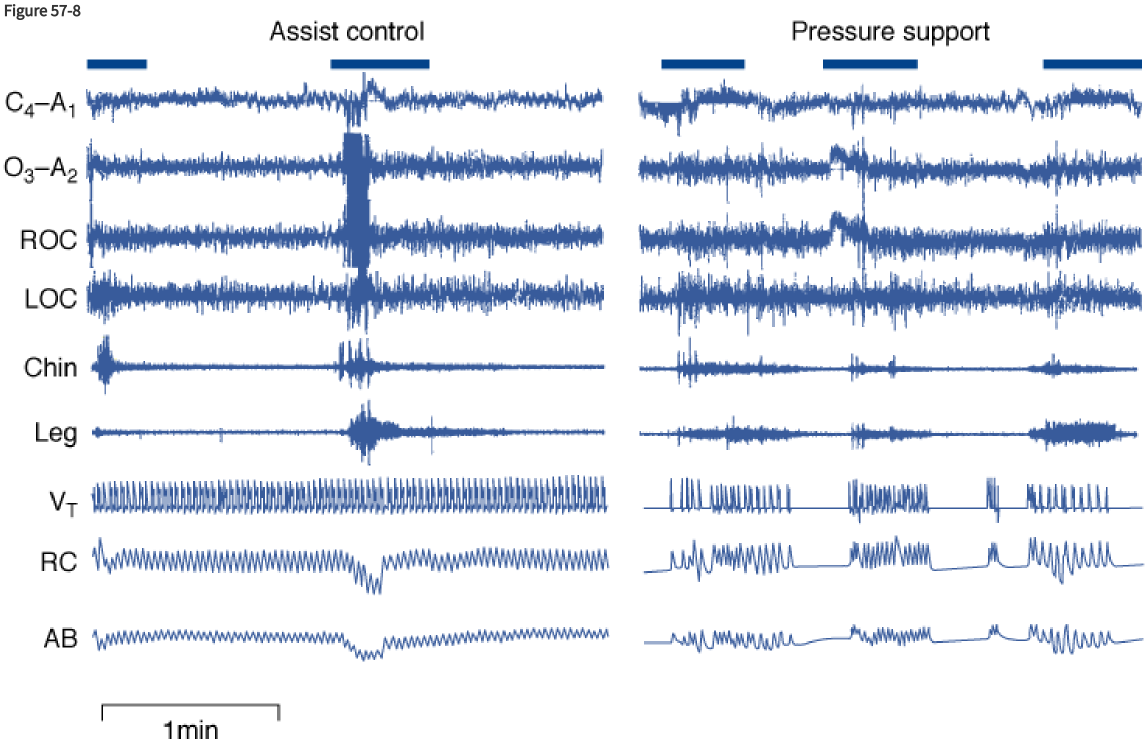
Mechanical Ventilation

In addition to the noise associated with ventilators and their alarms, mechanical ventilation can disrupt sleep by producing periodic breathing with recurrent central apneas, or through the development of patient–ventilator dyssynchrony. Recurrent central apneas occur when the ventilator assist is excessive relative to the patient's ventilatory demand.[52] This is most likely to occur during sleep when demand is lowest and has been observed in critically ill patients.[53]

Dyssynchrony occurs when cycling of the ventilator is not in phase with the patient's efforts. Thus, the ventilator may be delivering gas when the patient wants to exhale, and vice versa. This dyssynchrony is common during conventional mechanical ventilation. Leung et al found that 28% of patients' inspiratory efforts occur during the ventilator's expiratory phase (ineffective efforts).[54] When one considers that ineffective efforts are the most extreme form of dyssynchrony, the incidence of less extreme forms of dyssynchrony (e.g., excessive delay in triggering or in cycling off the ventilator) may be even more common. In awake individuals, such dyssynchrony is very uncomfortable. Thus, it would seem reasonable to expect dyssynchrony to result in arousals from sleep. Subjectively, 30% of patients in an ICU reported "agony/panic" during mechanical ventilation, which was associated with difficulties to synchronize the patients' breathing with the ventilator.[55]

In one of the first studies that investigated the impact of ventilator mode on sleep in the ICU, Parthasarathy and Tobin[56] reported a higher frequency of arousals and awakenings during pressure-support ventilation versus assist-control ventilation (79 ± 7 vs. 54 ± 7 per hour) in eleven critically ill patients (Fig. 57-8). Six patients, with underlying heart failure, had a high frequency of central apnea (53 ± 8 per hour) during pressure support, which was reduced (to 4 ± 2 per hour) by the addition of dead space (which eliminated the central apneas by increasing ventilatory demand). The decrease in central apneas was accompanied by a fall in the frequency of arousals and awakenings (83 ± 12 to 44 ± 6 per hour). The very high frequency of arousals and awakenings, particularly during pressure support, supports the notion that patient–ventilator dyssynchrony is an important source of sleep disruption in the ICU, and that it can be influenced by the mode of mechanical ventilation.

Figure 57-8



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Sleep disruption during mechanical ventilation. Polysomnographic tracings during assist-control ventilation and pressure-support ventilation in a single patient. Electroencephalogram (*C4 − A1,O3 − A2*), electrooculogram (*ROC,LOC*), electromyograms (chin and leg), integrated tidal volume (V_T), and rib cage (*RC*) and abdominal (*AB*) excursions on respiratory inductance plethysmography. Arousals and awakenings, indicated by *horizontal bars,* were more numerous during pressure-support than during assist-control ventilation. (Reproduced, with permission, of the American Thoracic Society. Copyright © 2002 American Thoracic Socieity. Parthasarathy S, Tobin MJ, 2002, Effect of ventilator mode on sleep quality in critically ill patients. *Am J Respir Crit Care Med.* 2002;166:1423–1429. Official Journal of the American Thoracic Society.)

Potential Consequences of Sleep Disruption

Sleep is a biologic requirement for survival, which has been well documented in laboratory animals.[57,58] Prolonged sleep-deprivation experiments (5 to 9 days) in healthy subjects have produced marked cognitive impairment,[59] some objective neurologic signs,[60,61] and collapse requiring hospitalization.[62] In addition, there are five specific areas in which sleep disruption can adversely affect the clinical outcome of patients in the ICU.

Intensive Care Unit Psychosis

This refers to a state of delirium that characteristically starts after admission to the ICU.[63,64] Delirium is estimated to occur in 19% to 32% of patients in the ICU.[63,65] The pathogenesis is multifactorial, but includes sleep disruption associated with the patient's acute illness.[66,67] In addition, the rebound of REM sleep referred to earlier can be associated with delirium and nightmares. Persistence of this altered mental state impairs the ability of patients to interact with their caregivers, and can increase morbidity and delay weaning from the ventilator and discharge from the ICU.[64,66,68]

Rebound of Rapid Eye Movement Sleep

During REM sleep, heart rate, respiration, and blood pressure are highly variable[1] and can exhibit transient irregularities, making REM sleep, and more specifically REM rebound, potentially dangerous to a recuperating critically ill patient. It has been hypothesized that REM rebound, with associated episodes of hypoxemia, variable blood pressure, and cardiac arrhythmias, may play a role in the etiology of postoperative myocardial ischemia and infarction.[15,18,19] This is supported by the observation that REM rebound peaks at approximately the same time that these delayed complications of anesthesia and surgery occur most frequently.[18] This cardiovascular instability has particular relevance in patients who have significant vascular comorbidities.

Delayed Weaning from Mechanical Ventilation

Studies in healthy volunteers and animal models show that sleep deprivation can cause negative nitrogen balance,[57,69] decreased respiratory muscle endurance,[70] and a blunted genioglossus electromyogram response to carbon dioxide, which becomes more severe with advancing age.[71] Other investigators report decreased ventilatory responsiveness to hypoxemia and hypercapnia[72–74] and increased upper airway compliance[75] following sleep deprivation. The combination of decreased central drive and reduced respiratory muscle endurance can be further aggravated by the fact that sleep deprivation is associated with increased oxygen consumption and carbon dioxide production.[76] Weaning difficulty is related to inability of respiratory muscles to cope with the demands imposed by ventilatory requirements and mechanical load.[77–79] Accordingly, the consequences of sleep disruption (decreased respiratory muscle endurance and increased metabolic rate, and, hence, ventilatory requirements) may delay weaning, particularly in patients who have limited respiratory reserve.

Host Defense

Studies in animals indicate that sleep deprivation leads to failure of host defense.[80] During 3 weeks of sleep deprivation, rats developed a progressive negative energy balance and sympathetic activation.[58,81] This culminated in host-defense failure manifested by bloodstream infection and a cachectic-like moribund state.[80] Sleep deprivation in rats is also associated with early infection of the mesenteric lymph nodes by both aerobic and facultative anaerobic intestinal bacteria with migration to major organs.[82] The translocation of bacteria and endotoxin from the intestinal mucosa and their dissemination to extraintestinal sites is thought to drive a systemic inflammatory state and multiorgan failure in critically ill patients.[83–86] The contribution of sleep disruption to this syndrome in patients has not been determined, although sleep loss in humans is associated with changes in some parameters of host defense.[87–89]

Chronic Insomnia

Chronic insomnia has been reported in a subset of acute respiratory distress syndrome survivors 6 months or more following discharge from hospital.[90] None of these patients had insomnia before their hospitalization, which raises the possibility that it was related to their acute illness. This hypothesis, however, was not supported by a subsequent study of 497 patients who required admission to the ICU and whose sleep quality was assessed by questionnaire before their hospitalization, and again 6 and 12 months after discharge.[91] Compared to a community-based control group, these patients were more likely to have disturbed sleep (odds ratio [OR] 3.62, 95% confidence interval [CI] 2.93 to 4.47). Disturbed sleep, however, was strongly associated with concurrent illness (OR 3.29, 95% CI 2.80 to 3.88) and not with indices of their ICU admission, such as diagnosis, APACHE, and length of stay. Moreover, in previously healthy patients, sleep improved over 6 to 12 months after discharge from hospital, in contrast to patients with concurrent disease whose sleep quality remained the same. These results imply that the sleep disturbance associated with critical illness resolves over time and whatever remains is predominantly caused by concurrent disease.

Strategies to Improve Sleep in the Intensive Care Unit

No studies have systematically evaluated strategies to improve sleep in the ICU. Nevertheless, several options can be considered based on our current understanding of the pathogenesis of sleep disruption in this patient population and on anecdotal experience.

Treatment of the Underlying Illness

All sources of sleep disruption related to a patient's underlying medical condition should be reduced as much as possible. This includes management of the acute illness, such as optimal control of postoperative pain, as well as maintaining preexisting treatment for chronic medical disorders such as chronic cough associated with chronic obstructive pulmonary disease or restless legs syndrome. Detailed management of all such conditions is beyond the scope of this chapter.

Optimization of the Intensive Care Unit Environment

Although this has the greatest intuitive appeal to improve sleep in the ICU, it has not been shown to have a dramatic effect. Studies that minimized disruption from light, noise, and nursing interventions did not decrease sleep disruption.[16] Successful reduction of ICU noise and light levels was not associated with improved sleep quality as assessed by the nursing staff.[92] The use of earplugs subjectively improved sleep in a group of acutely ill patients compared to a control group.[93] When healthy volunteers were exposed to recorded ICU noise in a sleep laboratory,[94] subjects without earplugs displayed an increased number of awakenings and decreased REM sleep, similar to previous studies.[44,45,49] More recently, white noise has been shown to reduce arousals during exposure to recorded ICU noise in the sleep laboratory;[95] however, this has not been studied in the ICU. Overall, modification of the ICU environment yielded modest improvement in sleep quality, which is consistent with the observation that the ICU environment is not the major source of sleep disruption.[51] Nevertheless, it is important that all reasonable measures be taken to avoid an excessively disruptive ICU environment.

Several nonpharmacologic therapies have been used successfully in the management of chronic insomnia.[96,97] These therapies can be applied to patients in the ICU, despite the fact they have not been evaluated in this population. Reestablishment of a regular sleep schedule and avoidance of excessive sleep during the day may be feasible in ventilated patients once they have recovered from their acute illness and are being weaned from the ventilator.

Medication

Patients in the ICU receive multiple medications that should be reviewed for their potential to cause sleep disruption or sleep loss (see Table 57-1). Medication can also be used to consolidate sleep, either a pure hypnotic, such as zopiclone, or medication that treats the underlying cause of sleep disruption such as anxiety, depression, or delirium. Few studies have addressed the impact of such interventions on sleep in critically ill patients. One study randomized ICU patients to nocturnal infusions of either midazolam or propofol.[98] Although sleep was not assessed objectively by polysomnography, daily self-assessment questionnaires showed improved sleep with both agents. An alternative approach by Shilo et al[37] found that patients on the hospital ward had the typical nocturnal peak in melatonin levels, whereas nocturnal secretion was blunted in ICU patients. Administration of melatonin to nonventilated ICU patients improved the duration and continuity of sleep, as assessed indirectly by actigraphy.[99] It is possible that other strategies to resynchronize circadian rhythm, such as bright-light therapy, which has been used successfully in ambulatory patients,[42] may improve sleep in some ICU patients.

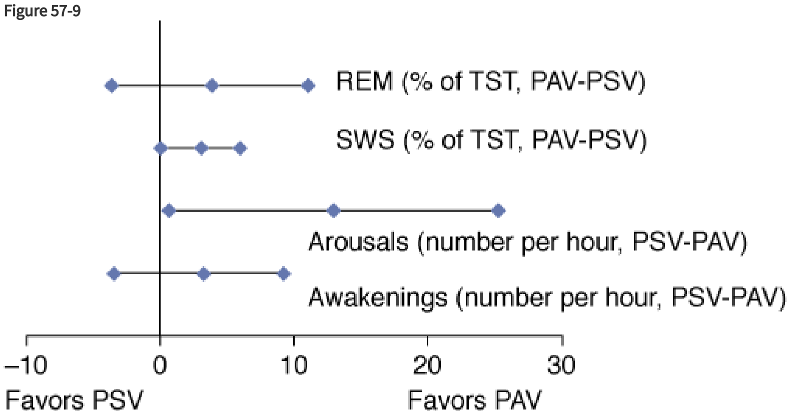
Mode of Mechanical Ventilation

The study by Parthasarathy and Tobin[56] quoted earlier supports the notion that some modes of mechanical ventilation are more "sleep-friendly" than others. Although assist-control ventilation is not free of dyssynchrony,[54] and may contribute to sleep disruption, they found that sleep fragmentation was less during assist-control ventilation than with pressure support, predominantly because of a reduction in the frequency of central apnea. Central apnea may occur either secondary to underlying disease, such as heart failure, excessive ventilator assistance, or both.[53]

In a randomized crossover study of twenty patients with chronic lung disease who were ready for extubation, Toublanc et al[100] compared polysomnography during assist-control ventilation and pressure support. Pressure support was set at a low level, 6 cm H_2O, whereas assist-control ventilation was titrated "until complete disappearance of spontaneous inspiratory efforts." Sleep architecture was superior during assist-control ventilation than with pressure support, as reflected by reduced wakefulness and increased NREM sleep. A further study compared assist-control ventilation, clinically adjusted pressure support, and automatically adjusted pressure support in fifteen mechanically ventilated patients and found no differences in sleep fragmentation across the three modes.[101] This may reflect appropriate adjustment of ventilator settings because minute ventilation was similar in all three groups of patients. Indeed, in patients with neuromuscular disease, Fanfulla et al[102] demonstrated that tailoring pressure support to the patient's respiratory effort resulted in improved sleep architecture by reducing the frequency of ineffective efforts.

Proportional-assist ventilation is a mode of ventilation that provides ventilatory support in proportion to the patient's inspiratory effort,[103] thereby optimizing patient–ventilator synchrony. The impact of pressure support and proportional assist ventilation on sleep was compared in thirteen patients as they were being weaned from mechanical ventilation.[104] Ventilator settings for both modes were titrated during a spontaneous breathing trial to reduce inspiratory work by 50%. The frequency of patient–ventilator dyssynchrony events was lower with proportional-assist ventilation than with pressure-support ventilation (24 ± 15 vs. 53 ± 59 per hour, $p = 0.02$), which resulted in a lower number of arousals (9, range: 1 to 41 vs. 16, range: 2 to 74) and improved sleep architecture (Fig. 57-9). In summary, the impact of mechanical ventilation on sleep depends upon the underlying disease, the stability of the control of breathing, the mode of mechanical ventilation and the ventilator settings.

Figure 57-9



Multivariate analysis of variance of the effect of modes of ventilation on comprehensive sleep quality. Central diamond indicates the mean absolute difference between proportional-assist ventilation (*PAV*) and pressure-support ventilation (*PSV*); *horizontal bar* indicates the 95% confidence interval. The combined effect of fewer arousals per hour, fewer awakenings per hour, greater rapid eye movement (*REM*) sleep, and greater slow-wave sleep (*SWS*) accounted for the significant ($p < 0.05$) overall improvement in sleep quality with PAV. *TST,* total sleep time. (Used, with permission, from Bosma et al.[104])

Important Unknowns

Despite increasing interest and research on sleep disruption in ventilator-supported patients, there is much that we do not know. We do not have a comprehensive understanding of the pathogenesis of sleep disruption in this patient population and the relative contribution of all of the potential sources of sleep disruption discussed above. We also do not know how sleep disruption changes over time in the ICU. Second, we do not know the impact of sleep disruption on outcomes such as weaning, length of stay in the ICU, and hospital morbidity and mortality. Third, it is not clear how much sleep can be improved in ICU patients and what treatment strategies work best.

The Future

Two issues need to be addressed to facilitate further research on this topic. First, the methodology to monitor sleep in the ventilator-supported patient needs to be addressed. Although attended polysomnography is the reference standard for monitoring sleep, it is labor intensive, costly, cumbersome for staff and patients, and not suited to repeated measurement, especially in the ICU environment. Alternative methodologies that address these limitations and are validated in mechanically ventilated patients are required. Second, larger studies are needed. To date, most studies have consisted of small numbers of patients from a single academic center. Although they have provided new and valuable data, their small sample size limits their applicability to the general ICU population and their ability to measure important clinical outcomes. These concerns could be addressed by multicenter studies that recruit larger numbers of patients.

Summary and Conclusions

There is strong evidence that sleep is severely disrupted in mechanically ventilated patients in the ICU, which has the potential to increase their morbidity and mortality through its effect on neurocognitive, cardiorespiratory, and immune function. Although sleep disruption has been attributed predominantly to the ICU environment, noise and patient-care activities account for less than 30% of sleep disruption.

Further research is required to provide a comprehensive understanding of the causes of sleep disruption in this patient population, which will form the basis for the evaluation of therapeutic strategies and their impact on both short-term and long-term clinical outcomes.

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